JAKHONGIR NODIROV

Robotics / Computer Vision Engineer

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SUMMARY

Computer Vision / AI engineer with hands-on robotics experience – ROS2 navigation and SLAM, sensor integration (LiDAR, encoders, motor control), and embedded deployment on Jetson and microcontroller platforms. Full-cycle experience from perception model research through real-time edge inference, including production deployment of CNN-based vision systems on embedded hardware. Recent work covers robot learning with LeRobot and latency analysis of a vision-language-action model. Author of 2 peer-reviewed publications on 3D medical imaging. Comfortable across perception, embedded control, and on-device optimization.

TECHNICAL SKILLS

- **Robotics:** ROS2, MicroROS, SLAM, Nav2, sensor integration (LiDAR, encoders, motor drivers), PID motor control
- **Robot Learning:** LeRobot, behavior cloning, PPO (Stable-Baselines3), MuJoCo
- **Embedded/Edge:** Jetson Nano, Jetson Xavier NX, Raspberry Pi, ESP32, TensorRT, ONNX, ONNX Runtime (C++), Yocto, PlatformIO, Arduino/C++
- **Perception & ML:** PyTorch, PyTorch3D, TensorFlow, OpenCV, Object Detection/Segmentation/Tracking, Pose & Landmark Estimation
- **Languages:** Python, C/C++
- **Deployment/Infra:** Docker/Compose, AWS EC2/S3, Oracle Cloud, CI/CD

WORK EXPERIENCE

HyVision System Inc. Seongnam, South Korea
AI Research Software Engineer Jun 2025 – Sep 2026

- Owned the full model lifecycle – dataset design, training, optimization, and on-device deployment – for CNN-based vision systems deployed on embedded/edge hardware in manufacturing equipment.
- Deployed trained models for real-time inference on production hardware using TensorRT/ONNX edge optimization.
- Worked directly with equipment integration, translating perception model output into reliable, low-latency decisions on physical machines.

AIVAR Seoul, South Korea
AI Research Software Engineer Feb 2023 – Mar 2025

- Led development of a 3D human avatar reconstruction server application (3D morphable face/body models), deployed on AWS with Docker for consistent dev-to-production environments.
- Built and shipped a deep-learning image verification system (vector embedding search) for a government-backed digital IP-protection initiative.
- Researched and optimized CNN- and transformer-based vision models across multiple concurrent product lines; owned deployment and production monitoring.

Gachon University – CT Lab Seongnam, South Korea
Graduate Researcher Sep 2020 – Feb 2023

- Designed a novel attention-based 3D U-Net architecture for volumetric MRI tumor segmentation, published in Sensors (2022).
- Built VTK-based 3D volume reconstruction from MRI slices, published at IEEE ICTC (2021).
- Contributed to applied CV lab projects including road-condition assessment and license-plate recognition.

SELECTED PROJECTS

Semiconductor Wafer Defect Inspection *HyVision, 2025*

- Built a high-precision defect inspection pipeline for 8-inch and 12-inch silicon wafers using CNN-based object detection, segmentation, and anomaly detection models, deployed on industrial edge GPUs inside wafer inspection equipment.
- PyTorch, ONNX/TensorRT edge optimization · Deployed in production for continuous, high-precision defect detection with a low false-positive rate.

Consumer Electronics Surface Inspection *HyVision, 2025-2026*

- Built an end-to-end automated defect inspection pipeline (segmentation, detection, classification) for glass and metal enclosure parts, deployed on AOI machines in a tier-1 consumer electronics client's production line.
- PyTorch, ONNX/TensorRT edge optimization · Achieved fine-grained defect detection at production line speed with a low false-positive rate.

Autonomous Driving Robot (ROS2) *Side Project*

- Designed and built a mobile robot and its Gazebo twin running one ROS2 stack: a Jetson Xavier NX on ROS2 Humble in Docker (first built on a Jetson Nano B01 with ROS2 Foxy), an ESP32-S3 driving TB6612 motor drivers and streaming encoder and IMU data at 50 Hz, an RP-Lidar for range sensing, and an ESP32-CAM for vision.
- slam_toolbox builds maps from the real lidar, and AMCL localization and Nav2 navigate the Gazebo twin, built from measurements of the chassis; an EKF fuses wheel odometry and IMU, and a web dashboard handles teleoperation (roslib.js). Demo video

Camera-Only Docking *Side Project, 2026*

- Built the path from teleoperated demonstrations to a deployable policy: ROS2 bags into a LeRobot v3 dataset, a ResNet18 behavior-cloning policy exported to ONNX, and a C++ safety gate between every command source and the motors.
- Ran pre-registered paired trials against a classical servo: 16/30 against 27/30 from unseen start poses, and demonstrations recorded through the robot's lens fixed a lens failure (0/30 to 23/30). Simulation only so far.

SmolVLA Latency *Side Project, 2026*

- Exported the SmolVLA vision-language-action policy as three ONNX graphs driven by a C++17 runtime, reproducing LeRobot's actions to 6e-6 in FP32 on the real LIBERO checkpoint, and ran them on TensorRT on a Jetson Xavier NX: one action chunk in 1.0 s at FP32 and 0.33 s at FP16, with FP16 actions within the range of LeRobot's own BF16.

Quadruped Locomotion *Side Project, 2026*

- Trained a PPO velocity-tracking policy for a custom 12-DoF quadruped in MuJoCo, reaching 0.045 m/s tracking error against a 0.5 m/s command over a full 20 s episode with no falls; exported to ONNX.

Driver Monitoring System (DMS) *Lab Project*

- Built a driver-monitoring pipeline on Jetson Xavier NX with nine TensorRT FP16 engines for face detection, landmarks, head pose, and eye and mouth state, plus alert logic, running end to end at 16–30 FPS. Tested on recorded video; not yet validated in a vehicle.

Seamline *Open Source, 2026*

- Built a desktop app for segmentation on Electron and PyTorch: label images, train a model, review its proposals least certain first, and retrain on what was accepted. Remote training runs detached over SSH and survives lost connections; large images open through tile pyramids. Tested on VisA circuit boards.
- Built on mySegLib, a YAML-driven PyTorch segmentation library with DDP multi-GPU training, mixed precision, chunked inference for large images, and ONNX export. CI runs the full pipeline on synthetic data.

3D Human Avatar Reconstruction *AIVAR, 2023-2024*

- Converted 2D images into textured 3D face/body avatars using 3D morphable face and body model architectures, with diffusion-based image correction and neural texture-matching; deployed on AWS with Docker.

IP Protection – Image Verification *AIVAR, 2024*

- Built a vector-embedding image verification system for animation-character and logo IP protection using CNN-based face/feature recognition models; part of a government-backed digital IP-protection project.

Attention 3D U-Net for Brain Tumor Segmentation *CT Lab, 2021-2022*

- Designed a novel attention-based 3D U-Net with multiple skip connections for volumetric MRI tumor segmentation, reaching whole-tumor Dice 0.8974 on BraTS-2020. Published in Sensors (2022), cited 130+ times.

EDUCATION

Gachon University 2020.09 – 2022.08
Master's Degree, Computer Engineering – GPA 4.17/4.5

PUBLICATIONS

- "Attention 3D U-Net with Multiple Skip Connections for Segmentation of Brain Tumor Images." *Sensors* 22, no. 17 (2022): 6501.
- "3D Volume Reconstruction from MRI Slices based on VTK." In *2021 International Conference on Information and Communication Technology Convergence (ICTC)*, pp. 689–692. IEEE, 2021.

LANGUAGES

Uzbek (Native) · English (Proficient) · Korean (Intermediate)